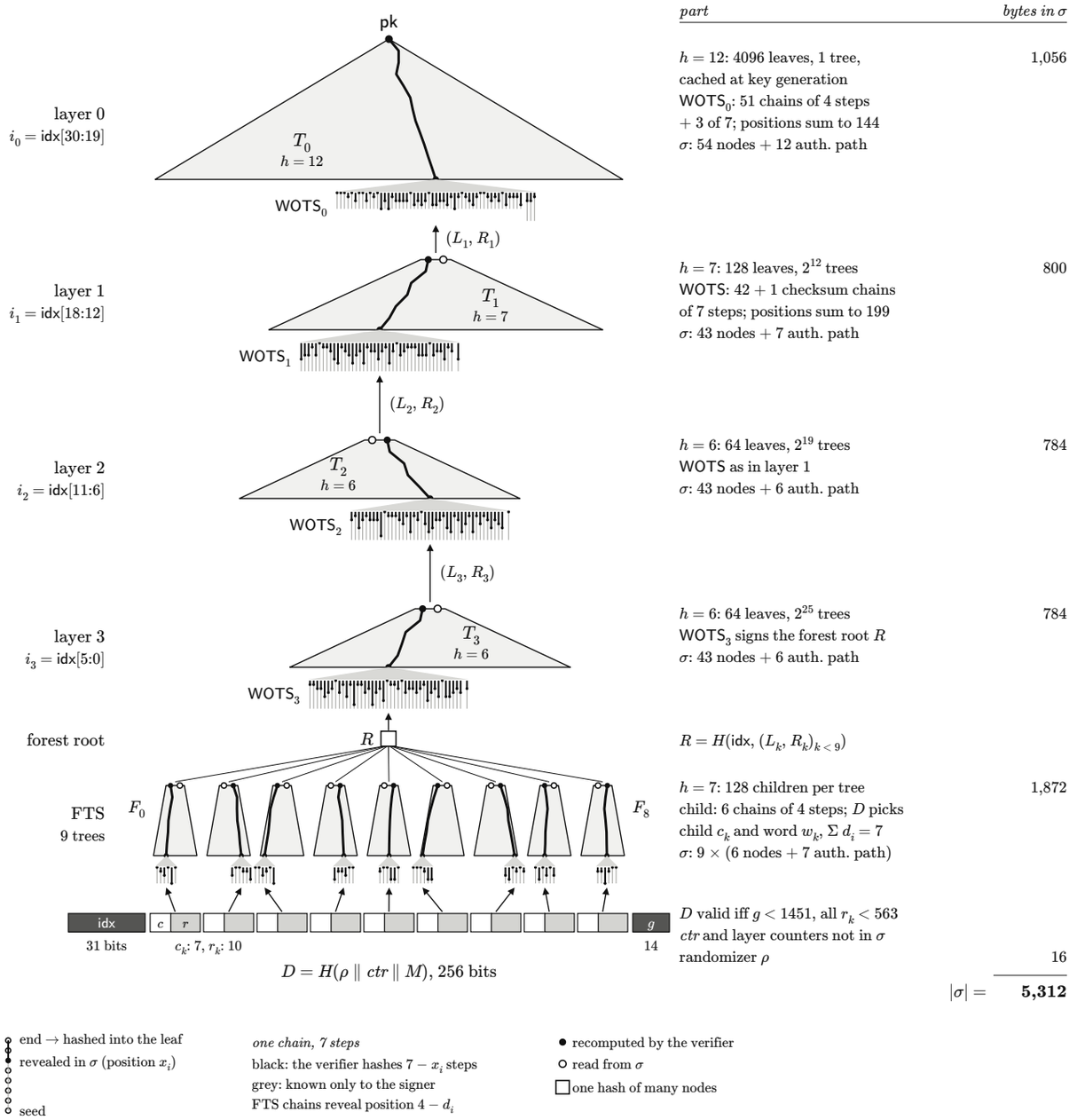


Fig. 1. Hypertree of submission a89db44d (sig.golf PR #819). Verification runs bottom-up. The digest D gives a 31-bit index idx and, for each FTS tree F_k, a child c_k and a word $w_k = (d_0, \ldots, d_5)$. The opened child reveals each of its 6 chains at position $4 - d_i$, so the verifier hashes $\Sigma\, d_i = 7$ steps per tree, then climbs to the tree's top pair (L_k, R_k). The 18 top nodes hash to the forest root R. In each layer, the message (R, or the top pair of the tree below) is hashed with a counter into chain positions x_i with a fixed sum (199, or 144 at the top); the verifier completes the chains, hashes the leaf and climbs to the next pair. Only the root of T_0 is computed; it must equal pk. The roots of T_1, T_2 and T_3 are never computed: the layer above signs their two top nodes directly, and R hashes the two top nodes of each F_k. Layers are numbered as in the Lean code (layer 0 is the top). The five grinding counters are not in σ: the expander finds them again and passes them to the verifier in the witness.

Table 1. Parameters of each part. Steps: hash steps from a chain's seed to its end. Sum: the fixed sum of the revealed chain positions. The two verify columns count what the verifier computes for one signature (the Lean *verify* in ClaudeWCT/WCT9/Core.lean). H is the contract's oracle: 64-byte blocks in, 32 bytes out, 8 cycles per block.

							verify		
part	h	*leaves*	*trees*	*leaf chosen by*	*chains × steps*	*sum*	*chain hashes*	*H calls*	*bytes in σ*
T_0 (top)	12	4,096	1	$\mathsf{idx}[30{:}19]$	$51 \times 4 + 3 \times 7$	144	81	95	1,056
T_1	7	128	2^{12}	$\mathsf{idx}[18{:}12]$	43×7	199	102	110	800
T_2	6	64	2^{19}	$\mathsf{idx}[11{:}6]$	43×7	199	102	109	784
T_3	6	64	2^{25}	$\mathsf{idx}[5{:}0]$	43×7	199	102	109	784
R (forest root)								1	
$F_0, \ldots, F_8$	7	128	$9 \cdot 2^{31}$	c_k in D	6×4 per child	17 per tree	63	126	1,872
D and ρ								1	16
total							450	551	5,312